

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

**Product Description:** Manganese powder  
**Cat No. :** S60345  
**CAS No** 7439-96-5  
**Molecular Formula** Mn

### Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### Details of the supplier of the safety data sheet

#### **Importer**

Fisher Scientific Korea  
D5,D6, Incheon Airport Logistics Complex  
150, Gonghangdong-Ro 296 Beon-Gil  
Jung-Gu, Incheon  
Tel: +82-1661-9555  
Fax: +82-2-2023-0603

#### **Supplier**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099

**E-mail address** Chem.KR@thermofisher.com

### Emergency Telephone Number

**Emergency telephone: Medical:** +(82) 070-7686-0086 or + 1-703-741-5970  
**CHEMTREC:** 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
**Korea:** 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

#### Physical hazards

Flammable solids

Category 2

#### Health hazards

Based on available data, the classification criteria are not met

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements

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## Signal Word

## Warning

### Hazard Statements

H228 - Flammable solid

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

### NFPA

Health  
2

Flammability  
2

Instability  
0

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component | Common Name              | CAS No    | Index No | Weight % |
|-----------|--------------------------|-----------|----------|----------|
| Manganese | No information available | 7439-96-5 | KE-22999 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

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**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Dry chemical.

**Extinguishing media which must not be used for safety reasons**

Water.

**Special hazards arising from the substance or mixture**

Combustible material.

**Hazardous Combustion Products**

Manganese oxides.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental precautions**

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

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## Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep under nitrogen.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | CAS No    | Korea                                                 | ACGIH TLV                                                 | OSHA PEL                                                                                                                                            |
|-----------|-----------|-------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Manganese | 7439-96-5 | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 mg/m <sup>3</sup> |

| Component | CAS No    | European Union                                                      | The United Kingdom                                                                                                                               | Germany                                                                                                                                                                                                                                                                                                                       |
|-----------|-----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manganese | 7439-96-5 | TWA: 0.2 mg/m <sup>3</sup> (8h) TWA:<br>0.05 mg/m <sup>3</sup> (8h) | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr | TWA: 0.2 mg/m <sup>3</sup> (8<br>Stunden). AGW - exposure<br>factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8<br>Stunden). AGW - exposure<br>factor 8<br>TWA: 0.2 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>TWA: 0.02 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>Höhepunkt: 1.6 mg/m <sup>3</sup><br>Höhepunkt: 0.16 mg/m <sup>3</sup> |

### ACGIH - Biological Exposure Indices

| Component | CAS No    | ACGIH - Biological Exposure Indices |
|-----------|-----------|-------------------------------------|
| Manganese | 7439-96-5 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Personal protective equipment

Use only those certified by the Korea Occupational Safety and Health Administration.

#### Respiratory Protection

No protective equipment is needed under normal use conditions

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## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Dark brown Powder Solid

Odor No information available

Odor Threshold No data available

pH Not applicable

Melting Point/Range 1260 °C / 2300 °F

Softening Point No data available

Boiling Point/Range 1900 °C / 3452 °F

Flash Point No information available Method - No information available

Evaporation Rate Not applicable Solid

Flammability (solid,gas) No information available

Explosion Limits No data available

Vapor Pressure No information available

Vapor Density Not applicable Solid

Specific Gravity / Density No data available

Bulk Density No data available

Water Solubility Insoluble

Solubility in other solvents No information available

### Partition Coefficient (n-octanol/water)

| Component | CAS No    | log Pow           |
|-----------|-----------|-------------------|
| Manganese | 7439-96-5 | No data available |

Autoignition Temperature No data available

Decomposition Temperature No data available

Viscosity Not applicable Solid

Explosive Properties No information available

Oxidizing Properties No information available

Molecular Formula Mn

Molecular Weight 54.94

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available

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## Chemical Stability

Moisture sensitive.

## Possibility of Hazardous Reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Exposure to moisture.

## Incompatible Materials

Acids. Bases. Halogens.

## Hazardous Decomposition Products

Manganese oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

**Inhalation** Avoid breathing dust or spray mist. May be harmful if inhaled.  
**Ingestion** May be harmful if swallowed.  
**Eyes** Avoid contact with eyes.  
**Skin** Avoid contact with skin.

#### Information on Health Hazards

##### (a) acute toxicity;

**Oral** Based on available data, the classification criteria are not met  
**Dermal** No data available  
**Inhalation** No data available

| Component | CAS No    | LD50 Oral             | LD50 Dermal       | LC50 Inhalation              |
|-----------|-----------|-----------------------|-------------------|------------------------------|
| Manganese | 7439-96-5 | LD50 = 9 g/kg ( Rat ) | No data available | LC50 > 5.14 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

**Respiratory** No data available  
**Skin** No data available

| Component | CAS No    | Test method       | Test species      | Study result      |
|-----------|-----------|-------------------|-------------------|-------------------|
| Manganese | 7439-96-5 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component | CAS No    | Test method       | Test species      | Study result      |
|-----------|-----------|-------------------|-------------------|-------------------|
| Manganese | 7439-96-5 | No data available | No data available | No data available |

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(f) carcinogenicity; No data available

| Component | CAS No    | Test method       | Test species / Duration | Study result      |
|-----------|-----------|-------------------|-------------------------|-------------------|
| Manganese | 7439-96-5 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|-----------|-----------|------------|------------|------------|------------|------------|
| Manganese | 7439-96-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component | CAS No    | Test method       | Test species / Duration | Study result      |
|-----------|-----------|-------------------|-------------------------|-------------------|
| Manganese | 7439-96-5 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

## Other Adverse Effects

No information available.

| Component | CAS No    | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-----------|-----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Manganese | 7439-96-5 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | CAS No    | Freshwater Fish                                         | Water Flea        | Freshwater Algae  | Microtox          |
|-----------|-----------|---------------------------------------------------------|-------------------|-------------------|-------------------|
| Manganese | 7439-96-5 | LC50: > 3.6 mg/L, 96h semi-static (Oncorhynchus mykiss) | No data available | No data available | No data available |

### Persistence and degradability

Persistence  
Degradability  
Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

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## Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

## Ozone Depletion Potential

| Component | CAS No    | Ozone Depletion Potential |
|-----------|-----------|---------------------------|
| Manganese | 7439-96-5 | Not listed                |

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN3089                          |
| Proper Shipping Name    | Metal powder, flammable, n.o.s. |
| Technical Shipping Name | Manganese                       |
| Hazard Class            | 4.1                             |
| Packing Group           | III                             |

### IATA

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN3089                          |
| Proper Shipping Name    | Metal powder, flammable, n.o.s. |
| Technical Shipping Name | Manganese                       |
| Hazard Class            | 4.1                             |
| Packing Group           | III                             |

### IMDG/IMO

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN3089                          |
| Proper Shipping Name    | Metal powder, flammable, n.o.s. |
| Technical Shipping Name | Manganese                       |
| Hazard Class            | 4.1                             |
| Packing Group           | III                             |
| Marine Pollutant        | No hazards identified           |

## **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

### **International Inventories**

ALFAAS60345

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| Component | CAS No    | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|-----------|-----------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Manganese | 7439-96-5 | KE-22999 | X    | 231-105-1 | X     | X   | -    | X     | X    |      | X    |

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Manganese | 7439-96-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|-----------|-----------|----------|------------------------------|---------------------------|
| Manganese | 7439-96-5 | Listed   | Not applicable               | Not applicable            |

## Korean National Regulations

| Component | CAS No    | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-----------|-----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Manganese | 7439-96-5 | Annex 1 - KE-22999                                                  | Not applicable       | Not applicable                              |

| Component | CAS No    | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-----------|-----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Manganese | 7439-96-5 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component | CAS No    | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-----------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Manganese | 7439-96-5 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component | CAS No    | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-----------|-----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Manganese | 7439-96-5 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component | CAS No    | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-----------|-----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Manganese | 7439-96-5 | Listed                                                       | Not applicable               | Not applicable                         |

| Component | CAS No    | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-----------|-----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Manganese | 7439-96-5 | Listed                               | Listed                                             | 1 mg/m <sup>3</sup> TWA            |

| Component | CAS No    | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals        | ISHA - Special management materials |
|-----------|-----------|-------------------------------------------------------------|-------------------------------------------------------|-------------------------------------|
| Manganese | 7439-96-5 | Not applicable                                              | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component | CAS No | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-----------|--------|----------------------------|---------------------------|--------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
|           |        |                            |                           |                                                              |                             |                                    |                             |

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|           |           |                |                                     |                            |                |                |                |
|-----------|-----------|----------------|-------------------------------------|----------------------------|----------------|----------------|----------------|
|           |           |                |                                     | <b>Substances When Wet</b> |                |                |                |
| Manganese | 7439-96-5 | Not applicable | 5. Metal powder<br>500<br>kg Powder | Not applicable             | Not applicable | Not applicable | Not applicable |

## Control Parameters

| Component | CAS No    | Korea                                                 | ACGIH - Biological Exposure Indices |
|-----------|-----------|-------------------------------------------------------|-------------------------------------|
| Manganese | 7439-96-5 | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component | CAS No    | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-----------|-----------|----------------------------------|----------------------------|
| Manganese | 7439-96-5 | Not applicable                   | Not applicable             |

CERCLA Not applicable

| Component | CAS No    | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-----------|-----------|-------------------------------------------|--------------------------|-------------------------------|
| Manganese | 7439-96-5 | Not applicable                            | Not applicable           | 1.0 %                         |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Warning.

H228 - Flammable solid.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

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## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

24-Nov-2010

## Revision Date

12-Jun-2024

## Revision Number

2

## Revision Summary

New emergency telephone response service provider.

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**